0580_w24_ms_22_Q9

Question	Answer	Marks	Partial Marks	Guidance
9	36π cao	**3**	**B2** for answer 113 or 113.0 to 113.1… or an answer in terms of π which rounds to 113 or **M1** for correct first step for finding *d* or *r* e.g. $72 + 72 = d^2$ oe $72 = r^2 + r^2$ oe or $\left(\frac{\sqrt{72}}{2}\right)^2 + \left(\frac{\sqrt{72}}{2}\right)^2 = r^2$ oe $\sqrt{72} = \frac{r}{sin45}$ oe $\frac{1}{2} \times r \times r = \frac{72}{4}$ oe	Ignore inaccurate values for 6 or 12 if recovered by accurate 36. B2 for: answers $6^2\pi$, $36\pi = 113$, 36π in the working and 36 as the answer e.g. $\sqrt{(\sqrt{72})^2 + (\sqrt{72})^2}, \sqrt{(\sqrt{72})^2 \times 2}, \sqrt{144}, d = 12$ $\sqrt{\left(\frac{\sqrt{72}}{2}\right)^2 + \left(\frac{\sqrt{72}}{2}\right)^2}, \sqrt{\left(\frac{\sqrt{72}}{2}\right)^2 \times 2}, r = 6$ $(\sqrt{72}) \times cos45$ $\sqrt{\frac{72}{2}}, \sqrt{36}, r^2 = \frac{72}{2}$ For $\sqrt{72}$ allow $6\sqrt{2}$ or 8.49 or 8.485… For $\frac{\sqrt{72}}{2}$ allow $3\sqrt{2}$ or 4.24 or 4.242 to 4.243